

ME597 IIOT Project Template

Directions: Please click "**File**" and select "**Save a copy in Drive.**" Then, share this with all of your teammates using their gmail addresses. Finally, please fill in your group number project title, and team members, below.

Group Number: 6

Project Title: Particle Monitoring System with Northrop Grumman

Team members: Robert Daniel Claud IV

Milestone 0: Project Charter (Complete by End of Week 4)

0.1 Initial Project Description

[Please provide a high level overview of the of project you hope to complete in this course. Please make sure to provide the following elements: The company you want to work with and the scope of the project]

For this course, I would like to collaborate with my company, Northrop Grumman, to install and collect useful data from particle, temperature, and humidity monitoring systems for use in a clean room environment. The scope of the project will be the following:

1. Establishing network connections with the sensors
2. Collecting data in a database
3. Building a live-feed dashboard for viewing of this data

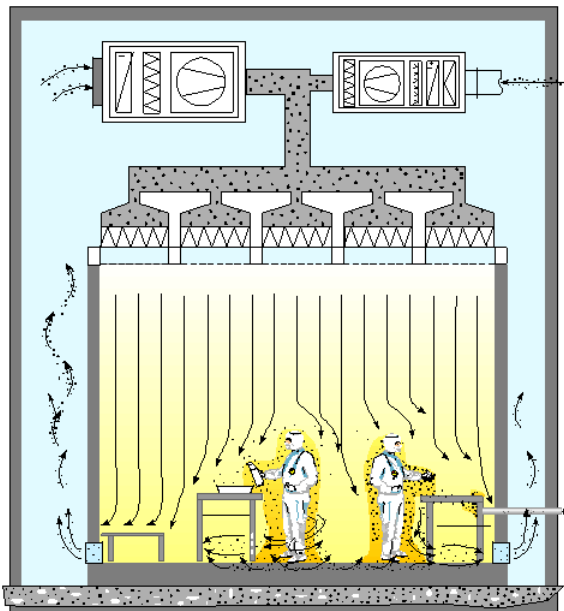
4. Developing AI models to predict when and where particle burst events will occur as well as out of spec temperature and humidity events.

The data collected from this project can be used to develop new strategies to minimize the occurrence of out of spec particle, temperature and humidity events.

0.2 Physical Asset

[Please provide a detailed description of the physical asset that you are going to monitor. Please provide pictures, figures, schematics, and any other relevant information you have at this time.]

The physical asset I will be monitoring is the ambient air inside of a cleanroom. This is a little non-standard since it is not a single piece of equipment, but rather an entire room. Cleanrooms are engineered rooms typically used in the pharmaceutical, semiconductor, space and healthcare industries. Cleanrooms are specifically engineered to minimize floating particulate matter in the air when airborne particles can impact the production of very small products. Cleanrooms are able to accomplish this using a series of airlocks, air filters, and by carefully controlling the pressure of the room. In addition, all objects inside of a cleanroom environment are selected to not produce dust or other particles. Various kinds of plastics are specifically concerning due to outgassing effects.



Example of a cleanroom taken from [Wikipedia](#)

0.3 Asset Role

[What role does it play in the big picture of the company?]

In the big picture of the company, the cleanroom is the manufacturing shop floor for Northrop Grumman's [microelectronics business](#). Every electronic device produced by the company is produced or tested within a cleanroom environment. Controlling particulate matter is critical as even the smallest particle can render a product defective. Cleanrooms are essential for guaranteeing the quality of the finished product and to ensure that none of the processes in manufacturing microelectronic devices are responsible for defective end products.

0.4 Company Benefits

[Please provide a brief explanation of how your project will provide a benefit to the company in question?]

This project will benefit the company by giving insights into what patterns (if any) exist for increased particle counts or out of spec temperature and humidity events. These insights can be used to change policies to minimize these events. For example, certain production areas within a cleanroom environment could be switched off while there are increased levels of particulate matter are detected, thus reducing scrapped product. This would introduce the cost savings of scrapping product as well as time savings of the time invested into investigating why scrap events occur.

0.5 Questions or Concerns?

[What questions or concerns do you have about this project that you would like feedback on?]

I would like further clarification on the amount of data that needs to be shared with the class. Since this project will be completed in collaboration with a defense contractor, much of the data is proprietary and is harder to get approved for distribution outside of the company. I would ask that the data and project presentations only be shared with US persons if possible.
